

# AI UX & Data Visualisation Design Principles (CA6002)

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## Chapter 4.3 – Finding AI Opportunities

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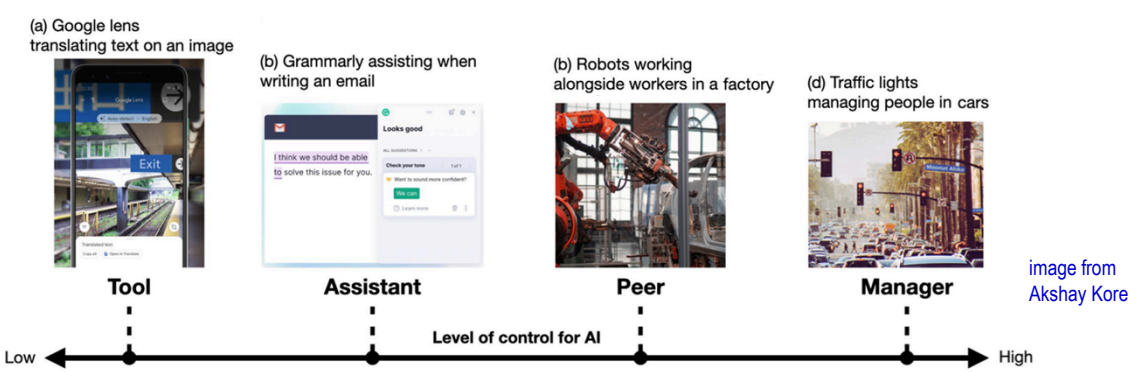
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# Finding AI Opportunities

## Roles for AI

- AI can be given different **levels of control** and these changes their **roles**. This can range from a tool (more control) to autonomously managing human activity (less control).

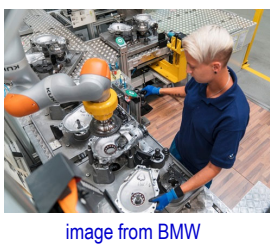


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# Finding AI Opportunities

## Roles for AI

- Tools** – AI as a tool provides the most amount of **human control** as the user can decide if they want to accept or ignore its output/recommendation (e.g. Google Lens provide language translation of text found in uploaded or live camera images).
- Assistants** – An AI assistant responds to the user request but is **more proactive** in taking context-sensitive action to assist (e.g. Grammarly will correct your grammar and also suggest context appropriate ideas to re-phrase your writing).
- Peers** – AI as a peer completes its own **independent function** without involvement of humans. Humans work along side such AI systems using a **cognitive division of labour** based on their respective strengths and capabilities (e.g. human collaborates with a robot in an axle drive assembly process in a car factory).



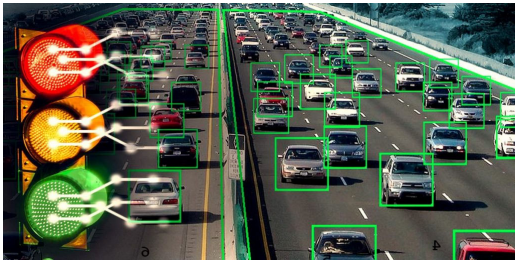
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## Finding AI Opportunities

### Roles for AI

- **Manager** – AI as a manager describes a role in which the AI system **organises human activities** or **assigns & coordinates** tasks for better outputs.
- Surrendering such excessive control may seem rather intimidating but this is a role we assign AI systems to function **autonomously** in order to managed human activities and production processes towards greater efficiency (e.g. AI monitors and adjust traffic light signals to optimise traffic flow)
- E.g. AI systems manage Alibaba’s automated packaging and warehousing for increase throughput.

AI system analyses live videos, adapts the lights’ timing to reduce traffic congestion.



[6] ELE Times News, AI Enabled Traffic Light System - <https://www.eletimes.com/ai-enabled-traffic-light-system>

Image from ELE Times News

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Think and Apply

AI Everywhere, AI Everyday

- Let us look at several commonly used applications and products with the goal of reflecting on **where**, **why** and **how** AI technologies have been embedded in them?

Online shopping

Google search

Smart fans



DeRUCCI smart mattress

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## Finding AI Opportunities

### Breaking Down Jobs into Tasks

- A **job** is a way of accomplishing a goal. A **task** is a **unit of activity** that a person performs to get their job done and most jobs consist of a set of multiple tasks
- To finding opportunities to deploy AI, first **identify jobs** that need to be done. Then break these jobs down into **individual tasks** that forms manageable chunks.

#### Run a full marathon with the help of a personal trainer

1. **Create training plan** – trainer to create a training plan for me based on my current level of fitness.
2. **Identify correct equipment** – find the proper shoes. The trainer can help me with this.
3. **Provide motivation** – this training will be long & gruelling. Trainer needs to push me to keep train.
4. **Reminder to stick to plan** – Trainer to remind me of the progressive workout routines & schedule.
5. **Inform of progress** – Trainer to give regular feedback on performance
6. **Update training plan** – Trainer need to update schedule and training plan if I miss some.



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

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## Finding AI Opportunities

### Mapping User Journey

- Once the job has been identified and broken down into manageable tasks, the next step is to create a **user journey** (i.e. task flow or workflow).
- Walking through a user journey can help us better understand what tasks **AI can help** (i.e. augment) and what can be entirely **delegated to an AI** system (i.e. automate).
- A **journey map** visualises the **steps** users go through within the product/service to achieve a goal and a realistic & useful journey map is built through user interviews, market research, contextual inquiry, direct observation, etc.
  - Journey maps a specific product/service & user persona in order to **identify pain points**.
  - Events are depicted chronologically, with the purpose of understanding user journey within product.
  - Map captures attributes like journey phases, actions, mindsets and emotions.
  - AI opportunities can be identified by thinking of journey phases as jobs and actions as tasks



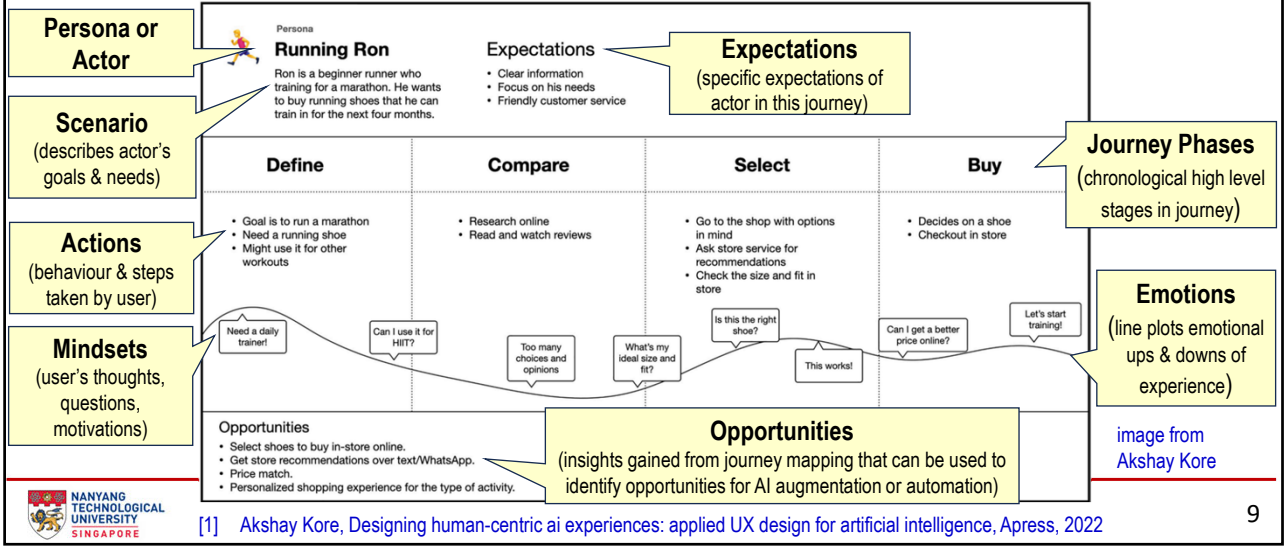
[8] Nielsen Norman Group, Journey mapping 101 - <https://www.nngroup.com/articles/journey-mapping-101/>

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Finding AI Opportunities

Journey Mapping - for a person buying a shoe in a store



Finding AI Opportunities

Problem Analysis - No sense to use AI

- After mapping the jobs and tasks into a user journey, the next step is to identify **which tasks** can be meaningful **enhanced by using AI** and which will not.
- Here are some cases where it makes no sense to use an AI solution:
  - **Errors can be costly** – in scenarios where the cost of errors outweighs the benefit that AI provides (e.g. the human cost of misdiagnosing a rare medical condition outweighs the productivity improvement attained through automated AI-powered diagnosis).
  - **Complete transparency** – in scenarios where precise and transparent explanations are needed to justify a critical decision (e.g. criminal sentencing). Its often hard to identify how a deep learning model arrives at a particular outcome.
  - **Limited data availability** – the quality of AI models often depends on large amounts of useful data being available for training. No data, no AI.

## Finding AI Opportunities

### Problem Analysis - No sense to use AI

- Some other reasons not to employ an AI solutions:
  - **Rapid time to market and low cost product** – building robust and trustworthy AI solution requires time for development and testing (e.g. building from scratch or with new data) and can be costly in terms of development resources (e.g. compute cost for model training, collecting lots of data, etc).
  - **Social interactions** – in scenarios where the product or service depends on high social intelligence (e.g. conversing & empathising with customers). AI systems are not good at interpreting subtle cues and non-verbal signals (e.g. eye wink or frown).
  - **People resistant to AI** – there are people who are resistant to dealing with AI tools as they may view them as too opaque, rigid, emotionless, independent and adversarial (but there are some strategies to help overcome this resistance).



[9] Julian De Freitas, Why People Resist Embracing AI, Harvard Business Review, Jan 2025 – <https://hbr.org/2025/01/why-people-resist-embracing-ai>

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## Finding AI Opportunities

### Problem Analysis - Make sense to use AI

- There are many scenarios that using AI can significantly improve the user experience, efficiency and quality of the outcome. Some of these cases are:
  - **Personalisation** – it is good to use AI to personalise the experience of different users (especially if user-relevant data profiles or usage behaviour are available for learning).
  - **Recommendation & ranking** – AI can surface recommendations that would be impossible for user to find on their own. With the large number of options, AI with **clear ranking logic** can determine the best options to recommend to the user.
  - **Recognition & categorisation** – AI systems can recognise a class of entities (e.g. faces) from large data samples (e.g. auto-tagging social media photos). It ability to categorised (i.e. cluster) entities into different sets can assist in automatic sorting (e.g. sorting defective products) and identifying groups (e.g. customer buying behaviour).



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

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## Finding AI Opportunities

### Problem Analysis - Make sense to use AI

- There are other use cases where using AI can be beneficial:
  - **Detecting anomalies** – AI systems are good at automatically identify unexpected changes in a data set’s normal behaviour (e.g. detecting anomalous spending pattern to flag out potential credit card fraud).
  - **Natural language understanding** – AI can be used to accurately translate language or transcribe speech to text. Voice-based assistant (e.g. Alexa or Siri) provide a seamless hand-free way to interact with AI systems.
  - **Generate new data** – Generative AI systems can assist in generating new multi-modal data (e.g. text, images, music, video) and to even do it in a particular style (e.g. paint in the style of Monet) after being trained with data examples in that style.



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

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## Finding AI Opportunities

### Problem Analysis - Identifying tasks suitable for AI

- Once the identified jobs and tasks have been mapped onto a user journey, the next step is to determine which of these tasks can be **meaningfully handled by AI**.

| Task name                  | Type of problem                 | Suitable for AI |
|----------------------------|---------------------------------|-----------------|
| Creating a training plan   | Personalization                 | Yes             |
| Finding equipment          | Recommendation, personalization | Yes             |
| Motivating the trainee     | Social intelligence             | No              |
| Sending workout reminders  | Recommendation, personalization | Yes             |
| Telling progress           | Prediction, anomaly detection   | Yes             |
| Updating the training plan | Personalization                 | Yes             |

image from Akshay Kore



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

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# Finding AI Opportunities

## Considerations for AI Tasks

- Figuring out suitable tasks is a critical part of finding AI opportunities but not all of these are **feasible** to be turned into AI solutions. Some considerations should be thought through before embarking on the AI project.
- **Data** – Good AI systems need large amount of representative data. Ask “*Is this data available?*”, “*Can I access it?*”, “*Do I have good quality data and are they accurately labelled?*”
- **Cost, time and effort** – Much resources are often needed to implement AI systems. Ask “*Is the cost (i.e. data collection, development manpower, compute cost, etc) of building & maintaining the AI system greater than the value it will brings?*”, “*Do we have the time & effort to train, test and deploy the AI system?*”, “*Do I have a justifiable value proposition for the AI solution (e.g. improved outcomes beyond human capabilities)?*”

# Finding AI Opportunities

## Considerations for AI Tasks - Augmentation

- An important consideration is to evaluate whether the AI should **automate** the task (i.e. complete task independently) or **augment** the human user by extending his ability to perform the task (e.g. Autodesk Generative Design tool produces alternative designs based on human given constraints).Types of tasks suitable for AI augmentation include:
  - **Personally valuable** – people enjoy the task (e.g. writing poetry) so the AI should not do it for them but support this activity (e.g. suggesting alternative words and phases that rhymes)
  - **Personal responsibility** – and empathy is needed (e.g. giving performance appraisal, sentencing ). AI can recommend an action but it requires human judgement to carry out the activity.
  - **High stakes** – when the outcome is critical (e.g. performing surgery) AI should assist surgeon with greater precision control (e.g. robotic arm) and enhanced information (e.g. 3D tumour outline overlay)
  - **Ill-defined** – some tasks are hard to define as they lack specifics (e.g. food taste, home interior design). Users can retain more control and the AI helps with recommending options (e.g. furnishing, colour, etc).

## Finding AI Opportunities

### Considerations for AI Tasks - Automation

- With automation, the AI systems performs the task **independently** and **without human** involvement. Types of tasks suitable for AI automation include:
  - **Boring and repetitive** – humans are often happy to delegate boring (e.g. categories 1000s of photos, computing tax returns) and repetitive (e.g. transcribing hours of interviews) tasks to an AI system.
  - **Low stakes** – task where occasional errors are inconsequential (e.g. recommending movies) relative to the benefits provided by the AI system most of the time (e.g. discovering new interesting movies).
  - **Dangerous and uncomfortable** – people prefer machines to do dangerous and painstaking tasks (e.g. high rise façade crack inspection) if AI-controlled machines (e.g. drones) are now capable alternatives.
  - **Superior to human performance** – there are tasks that AI systems outperform humans (e.g. real-time coordination, optimising resources, parsing large volume of data) due to their easy access to distributed data, computational speed and consistency. Such tasks should be automated so that the human user can focus on tasks they are good at (e.g. handling unstructured situation, social engagement)



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

## Finding AI Opportunities

### Considerations for AI Tasks - Augment vs Automate

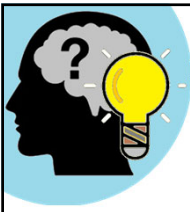
- After appropriate considerations, the type of action (i.e. augment or automate) for each tasks should be determined. The training for a marathon example is shown here.

| Task name              | Suitable for AI | Characteristics of task                          | Type of action          |
|------------------------|-----------------|--------------------------------------------------|-------------------------|
| Creating training plan | Yes             | Boring, lack of knowledge, tedious               | Automate                |
| Find equipment         | Yes             | Lack of knowledge; specifics hard to communicate | Automate and/or augment |
| Motivate the trainee   | No              | Not applicable                                   | Not applicable          |
| Send workout reminders | Yes             | Repetitive                                       | Automate                |
| Telling progress       | Yes             | Repetitive; specifics may be hard to communicate | Automate and/or augment |
| Update training plan   | Yes             | Specifics may be hard to communicate             | Augment or don't use AI |



[1] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

table from Akshay Kore



**Think and Apply**

**No AI, AI Autonomous or AI Augmented?**

- An exercise in finding opportunities for deploying AI solutions

**Weekly Meal Management**

- Breaking the job into smaller tasks.
- Defining what it mean to use an AI autonomous or an AI augmented approach to implement the task?
- Decide on the AI implementation approach for each task and describe how you would do it.





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**Finding AI Opportunities**

**Considerations for AI Tasks - Cultural traits**

- Statistics reveals **geographical differences** in the acceptance of AI. **26-36%** of **Western** (i.e. France, Spain, US) companies compared to **50-59%** of **Asian** (i.e. India, Singapore, China) reported to have actively deployed AI in their business operations.
- The **individualist** cultures (i.e. Western) are more prone to interpret AI features to **infringe** upon their **uniqueness**, **autonomy** and **privacy** compared to the **collectivist** ones (i.e. Asian), who tend to interpret AI features to facilitate conforming to consensus, respond to their environment and able to **protect** their privacy.
- Individualistic cultures (e.g. US) place a high value on **uniqueness** as it bolster their sense of self-reliance and being distinct from others. **Framing** AI technology as a way to **amplify** one's **uniqueness** may improve the individualist respond to AI (e.g. they are more accepting of personalised recommendation based on their unique preferences)



[11] A.J. Barnes et. al, AI and culture: Culturally dependent responses to AI systems - <https://www.sciencedirect.com/science/article/pii/S2352250X24000514>

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## Finding AI Opportunities

### Considerations for AI Tasks - Cultural traits

- **Individualists** tend to prescribe their actions as internally driven and independent from outside influence. Such cultural traits may influence the extent one perceive AI to **limit agency** (e.g. a study showed that such people had comparatively lower satisfaction using an AI voice assistant to order room service than a touch panel due to a lack of perceive control).
- **Algorithm aversion** is a phenomenon where people are reluctant to use AI algorithms after learning that they are imperfect (despite the fact the AI often predicts better then them). A study conducted in a Western cultural setting showed that people becomes **satisfied** even with a **slight control** over the process (e.g. given limited ability to modify inputs to the algorithmic decision making).
- Such cultural traits may influence the **willingness to disclose** (e.g. give opinions on discussion board, reviews) and express greater anxiety about **privacy disclosure** by AI apps (e.g. uncomfortable with personalised recommendation by AI apps like TikTok, Youtube, etc).



[12] A. Fan et. al, To talk or to touch: ., Feb 2022 - <https://www.sciencedirect.com/science/article/pii/S0278431921002553>  
[13] B.J. Dietvorst et. al, Overcoming Algorithm Aversion:.- <https://faculty.wharton.upenn.edu/wp-content/uploads/2016/08/Dietvorst-Simmons-Massey-2018.pdf>

## Finding AI Opportunities

### Summary

- Humans and AI systems have their respective strengths and weaknesses. Solutions that exploit their **synergistic collaboration** will yield the best outcome.
- The amount of control given to AI determines its **role** in the activity and finding **opportunities** to deploy AI solutions requires us to first break jobs down to tasks.
- Mapping the **user journey** helps visualise the **user's experience** across the tasks. This process gives a better understanding where AI can help by **augmenting** user's ability and where the task can be entirely delegated to AI **automation**.
- Appropriate **considerations** should be given to identify when it is appropriate and not appropriate to deploy an AI-based solution. These requires us to understand the **strength** and **weakness** of AI algorithms, **data** and **business** constraints, prevailing **cultural norms** where this AI solutions is being deployed.



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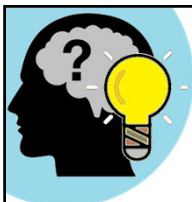
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**Think and Apply**

**AI Adoption across Different Cultures**

- Why is it that a super apps (e.g. WeChat) can be successful in China but not in the US?
- Do the sentiments towards AI at work vary across the different countries?
- Does age shape how confident one feels about using AI ?
- What do people really want from using AI?





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